

JANUARY – 2021

CS-15: RDMS USING ORACLE

Q-1(A). Answer the following. (04)

1. Give the full form of DCL - **Data Control Language**.
2. Describe title and repheader SQL *Plus formatting command.

A. The title command defines the top title.

Example: title center 'system user'

- ☒ The **repheader** command defines the report header.
- 3. Give the full form of DQL - **Data Query Language**.
- 4. Give the full form of SQL - **Structure Query Language**.

Q-1(B). Answer in Brief. (02)

1. What is DBMS?

A. DBMS stand for Database Management System

- ☒ DBMS used to manage your data like insert, update and delete etc.

Q-1(C) Answer in Detail. (03)

1. Write a note on Normalization.

- Normalization is a database design technique which organizes tables in a manner that reduces redundancy and dependency of data.

- ☒ It divides larger tables to smaller tables and links them using relationships.

- ☒ There are three types of rules.

☒ **1NF (First Normal Form) Rules**

- Each table cell should contain a single value.

- Each record needs to be unique.

☒ **2NF (First Normal Form) Rules**

- Rule 1- Be in 1NF
- Rule 2- Single Column Primary Key

☒ **3NF (First Normal Form) Rules**

- Rule 1- Be in 2NF
- Rule 2- Has no transitive functional dependencies.
- To move our 2NF table into 3NF, we again need to again divide our table.
- There are no transitive functional dependencies, and hence our table is in 3NF.
- In Table 3 Salutation ID is primary key, and in Table 1 Salutation ID is foreign to primary key in Table 3.

Q-1(D) Answer the following.

(05)

1. Difference between DBMS v/s RDBMS.

DBMS	RDBMS
Relationship between two tables or files are maintain programmatically.	Relationship between two tables or file can be specified at the time of table creation
Does not support client/server architecture.	Supports client/server architecture.
Does not support distributed database.	Supported distributed database.

No security of data.	Multiple security levels; 1. Login at OS level 2. Command level security 3. Object level security
Each table is given extension.	Many tables are grouped in one database in RDBMS.
Example: foxpro	Example: oracle

Q-2(A). Answer the following.

(04)

1. Give the full form of DML – **Data Manipulation Language**
 2. Give the full form of DDL – **Data Dictionary Language**
 3. Give the full form of RDBSM – Relational Database Management System
 4. Describe set linesize and set heading sql *plus formatting command.
- A. To avoid this problem, you can change the oracle default settings to increase width of the lines, using the set commands.
Example: set linesize 100
- ☒ The new heading will remain in effect until you enter different headings, reset each column's format, or exit from SQL *Plus.
Example: set heading on/off

Q-2(B). Answer the following.

(02)

1. **What is BCNF?**
- A. Boyce Code Normal Form
- ☒ Boyce Code Normal Form is based on functional dependencies that take into account all candidate keys in a relation.

- ⊗ However, BCNF also has additional constraints compared with the general definition of 3NF.

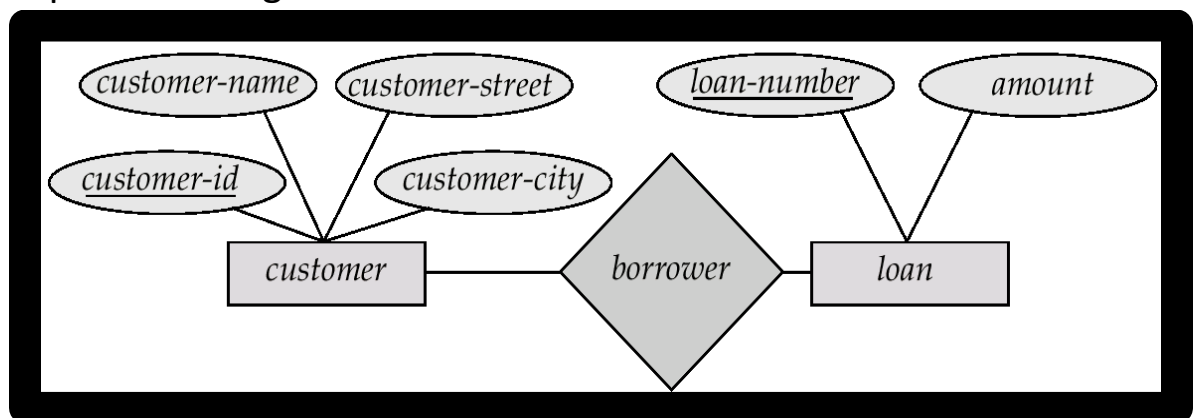
Rules for BCNF:

- ⊗ Rule 1: The table should be in 3rd normal form
- ⊗ Rule 2: X should be a superkey for every functional dependency (FD) $X \rightarrow Y$ in a given relation.

Q-2(C) Answer in Detail.

(03)

1. Explain E R Diagram.



- **Rectangles** represent table sets.
- **Diamonds** represent relationships between tables.
- **Lines** link attributes represent connection between tables with relationship and field.
- **Ellipses** represent table fields
- **Underline** indicates primary key attributes

Q-2(D) Answer the following.**(05)**

1. Difference between SQL v/s SQL *Plus.

SQL	SQL *Plus
SQL is a language for communicating with the oracle server to access data	SQL *PLUS recognizes SQL statements and sends them to Server
SQL is based on American National Standards Institute(ANSI)	SQL *PLUS is the Oracle proprietary interface for execution SQL statements
SQL manipulates data and table definition in the database	SQL *PLUS does not allow manipulation of values in the database
SQL is entered into the SQL buffer on one or more lines	SQL *PLUS is entered one line at a time, not stored in the SQL buffer
SQL does not have continuation character	SQL *PLUS uses a dash(-) as a continuation character if the command is longer than one line
SQL uses a termination character to execute commands immediately	SQL *PLUS does not require termination characters ; executes commands immediately

SQL uses functions to perform some formatting	SQL *PLUS uses commands to format data
---	--

Q.3 (A) Answer the following:-

1. A _____ is a database object that holds users data.

Ans. Table

2. All table columns belong to a single record.[True/False].

Ans. False

3. The update command is used to change or modifying data values in a table.[True/False]

Ans. True

4. A _____ keys represents relationship between tables.

Ans. Foreign keys

(B) Answer in brief

What is candidate key?

Ans.

A **candidate key** is a key that uniquely identifies rows in a table. Any of the identified candidate keys can be used as the table's primary key. Candidate keys that are not part of the primary key are called alternate keys.

One can describe a candidate key as a super key that contains only the minimum number of columns necessary to determine uniqueness.

(c) Answer in detail

Describe abs(), least(), months_between() functions.

□ abs(): returns absolute value

e.g:- Select abs(-12) from dual;

□ Least : returns smallest value out of list of nos.

e.g:- Select least(10,20,30) from dual;

□ MONTHS_BETWEEN(x,y) returns the number of months between x and y. If x is before y on the calendar, the number returned is positive; otherwise the number is negative.

e.g:- **SELECT** MONTHS_BETWEEN('25-MAY-2009',
'15-JAN-2012') **FROM** dual;

result:-

-31.677419

(D) Write a note on select statement.

The SELECT statement is a SQL statement that is used to retrieve data from a database. It is one of the most commonly used SQL statement

The syntax for a SELECT statement is as follows:

```
SELECT column_list FROM table_name  
[WHERE condition]
```

[ORDER BY column_name]

Column list

The column_list specifies the columns that you want to retrieve from the table. The * in the column_list indicates that all of the columns from the table should be retrieved.

Table name

The table_name specifies the name of the table that you want to retrieve data from.

WHERE condition

The WHERE clause is used to filter the rows that are returned. The condition can be a simple comparison, such as country = 'USA', or it can be a more complex expression.

ORDER BY clause

The ORDER BY clause is used to sort the rows that are returned. The column name that you specify in the ORDER BY clause determines the sort order. The default sort order is ascending, but you can also specify descending order by adding the DESC keyword after the column name.

Q4 (A) Answer the following:-

1. A _____ constraint must be specified as a logical expression that evaluate true or false.

Ans. Check

2. A _____ is a small Oracle Worktable, which consists of only one row and one column.

Ans. Dual

3. The _____ clause imposes a condition on the group by clause.

Ans. Having

4. A subquery is also termed as nested query. [true/false]

Ans. True

(B) Answer in brief:-

Describe initcap(), soundex() functions

□ Initcap() : converts first character of all words into capital and remaining into small

e.g:- Select initcap('welcome to hjd college') from dual;

☒ soundex():-

The SOUNDEX() function returns a string that contains the phonetic representation of a string.

Syntax:-

SOUNDEX(expression)

the The SOUNDEX() function is useful for comparing words that sound alike but spelled differently in English.

expression is a literal string or an expression that evaluates to a string.

The SOUNDEX() function will return a string, which consists of four characters, that represents the phonetic representation of the expression.

e.g:- SELECT
 SOUNDEX('see') see,
 SOUNDEX('sea') sea
FROM
 dual;

Result:-

SEE SEA

---- ----

S000 S000

Because both words sound the same, they should receive the same Soundex value.

(C) Explain GRANT, REVOKE , ROLLBACK command

□ **GRANT** : Grant is use to grant privileges on tables, view, procedure to other users or roles

e.g:-

-> grant select, update, insert on emp to scott;

-> grant all on emp to scott;

□ **REVOKE** :Use to take back privileges granted to other users and roles.

e.g:-

revoke select, update, insert on emp from scott;

□ **ROLLBACK** : the **ROLLBACK statement** is used to undo the work performed by the current transaction or a transaction that is in doubt.

Syntax:

Rollback [work] [to [savepoint] savepoint_name | force 'string']

Parameters or Arguments

WORK

Optional. It was added by Oracle to be SQL-compliant. Issuing the ROLLBACK with or without the WORK parameter will result in the same outcome.

TO SAVEPOINT savepoint_name

Optional. The ROLLBACK statement undoes all changes for the current session up to the savepoint specified by *savepoint_name*. If this clause is omitted, then all changes are undone.

FORCE 'string'

Optional. It is used to force the rollback of a transaction that may be corrupt or in doubt. With this clause, you specify the transaction ID in single quotes as *string*. You can find the transaction ID in the system view called DBA_2PC_PENDING.

e.g:-

□ Rollback to savepoint s1;

(D) Write a note on join.

A JOIN clause is used to combine rows from two or more tables, based on a related column between them.

Different Types of SQL JOINS:-

1.Inner Join

2. Outer Join

- a. Left outer Join
- b. Right outer Join
- c. Full outer Join

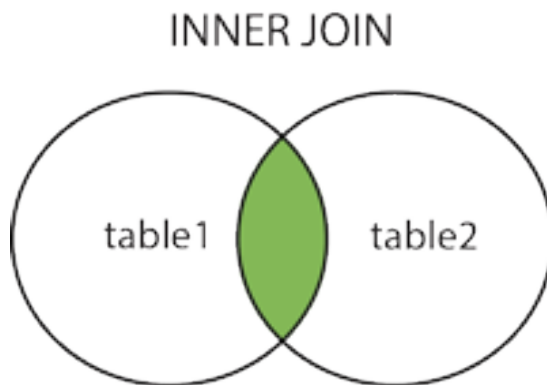
3. Self Join

4. Cross Join

1. SQL INNER JOIN Keyword

The INNER JOIN keyword selects records that have matching values in both tables.

This join is also known as equi join

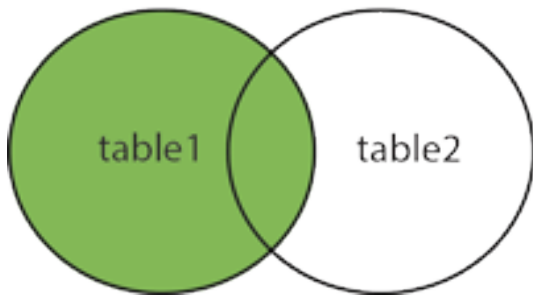


2. Outer Join

a. SQL LEFT JOIN Keyword

The LEFT JOIN keyword returns all records from the left table (table1), and the matched records from the right table (table2).

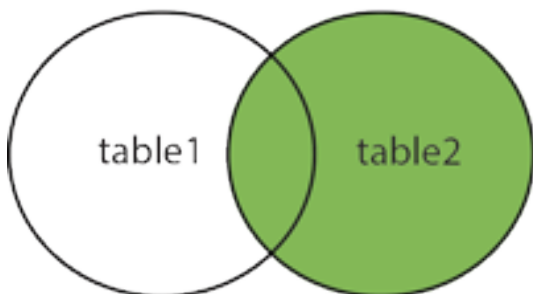
LEFT JOIN



b. SQL RIGHT JOIN Keyword

The RIGHT JOIN keyword returns all records from the right table (table2), and the matched records from the left table (table1).

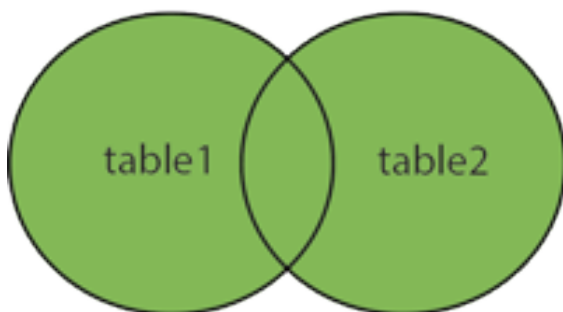
RIGHT JOIN



c. SQL FULL OUTER JOIN Keyword

The FULL OUTER JOIN keyword return all records when there is a match in either left (table1) or right (table2) table records.

FULL OUTER JOIN



3. SQL Self JOIN

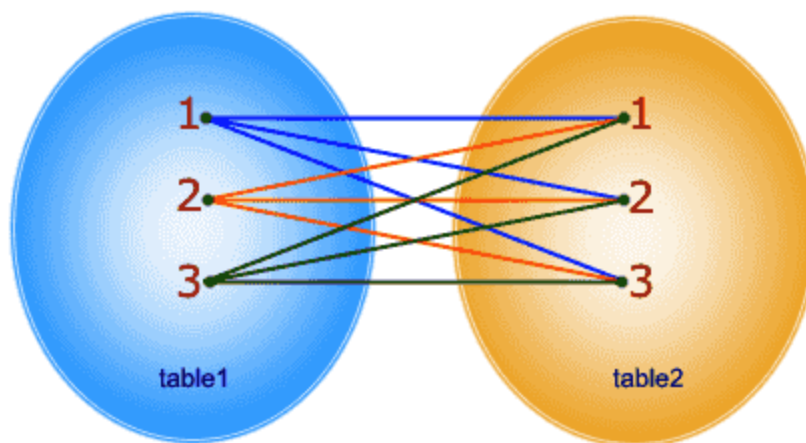
A self JOIN is a regular join, but the table is joined with itself.

4. Cross join

In this join there is no condition given so records multiply by each other table

Like if one table have 3 records and other table have 4 and we apply cross join than result will be 12 rows display

```
SELECT * FROM table1 CROSS JOIN table2;
```



In CROSS JOIN, each row from 1st table joins with all the rows of another table.
If 1st table contain x rows and y rows in 2nd one the result set will be $x * y$ rows.

Q-5 (A) Answer the following

(04)

- 1) A view is used to only look and table data and nothing else. The view is called a **Read Only View**.
- 2) In a **B-Tree** index, you walk the branches until you get to the node that has the data you want to use.
- 3) A **Database Link** object is telling oracle how to get from one database to another.
- 4) Two database operations wait for each other to release a lock is known as **Deadlock**.

(B) Answer in brief.

(02)

1) What are latches?

- Ans.**
- > A latch is a low-level internal lock used by oracle to protect memory structure.
 - > Latches are serializations mechanisms that protect areas of oracle's memory.
 - > The latches within oracle can be broken down into short and long wait latches.
 - > The purpose of latches is to manage simultaneous access to shared data structures so that only one process can access the structure at a time.

(C) Answer in detail.

(03)

1) Explain cluster.

- Ans.**
- > Clusters in SQL are used to store data that is from different tables in the same physical data blocks.
 - > They are used if records from those tables are frequently queried together.
 - > Each cluster stores tables data and maintains a clustered index to sort data.
 - > Columns within the cluster index are called clustered keys.
 - > These determine the physical placement of rows within the cluster.
 - > Cluster key is usually a foreign key of one table that references the primary key of another table in cluster.
 - > The first cluster is created. Then cluster index is created on cluster key columns.
 - > After cluster key index is created, data can be entered into tables stored in clusters.

=> Advantages of clusters :

- > Disk I/O is reduced.
- > Access time improves for joins of clustered tables.

=> Disadvantages of clusters:

- > Columns that are often updated are not good candidates for cluster key.
- > Reduces performance of INSERT statements as compared to storing the table separately with it's own index.

EXAMPLE:

```
create table branch_master  
(branch_code varchar(10) PRIMARY KEY,  
branch_name varchar(15) );
```

-> Now, Consider the above created table and let's create the cluster for the same.

Creating a Cluster :

```
create cluster branch_info  
(branch_code varchar(10) );
```

(D) Answer the following. (05)

1) Write a note on sequence.

Ans. -> A sequence is a set of integers that are generated in order on demand.

-> Sequences are frequently used in databases because many applications require each row in a table to contain a unique value and sequences provide an easy way to generate them.

-> Sequences can be created, changed, and dropped with the following three SQL commands:

- 1) CREATE SEQUENCE
- 2) ALTER SEQUENCE
- 3) DROP SEQUENCE

Syntax:-

Create sequence <name>

[Start with no

Increment by

Minvalue / Nominvalue

Maxvalue / Nomaxvalue

Cycle / Nocycle

Cache / Nocache]

->A sequence definition may consist of a start value, increment value, minimum value, and maximum value.

-> All sequence attributes are optional and they all have default values.

-> Each sequence has two columns: NEXTVAL and CURRVAL.

-> NEXTVAL is use for nextvalue in sequence while CURRVAL use for current value of data.

EXAMPLE:

-----create sequence-----

Create sequence stud_seq

Start with 10

Increment by 2

Minvalue 5

Maxvalue 30

Cycle

Cache 2;

-----create table-----

Create table stud

(sno number(3),
Sname char(30),
Seqno number(3),
Gender char(10));

-----insert records-----

Insert into stud values(&sno,'&sname',stud_seq.nextval,'&Gender');

Q-6 (A) Answer the following. (04)

- 1) Oracle used a method called **Multi-version concurrency control (MVCC)** to implement concurrency control.
- 2) The oracle engine automatically locks table data while executing SQL statement. [true/false]

Ans. -> True

- 3) The select.. for update can be used with set operators and group by clause. [true/false]

Ans. -> False

- 4) Latches request can be made two modes "**willing to wait**" or "**no wait**".

(B) Answer in brief. (02)

- 1) what is synonym?

Ans. -> A synonym is an alternative name for objects such as tables, views, sequences, stored procedures, and other database objects.

-> Synonyms is dependent object.

-> A synonym is an alias for a schema object.

SYNTAX:

Create Synonym Synonym_Name for Table_name;

(C) Answer in detail.

(03)

1) Explain Deadlock.

Ans. -> A deadlock occurs when two or more sessions are waiting for data locked by each other, resulting in all the sessions being blocked.

-> Oracle automatically detects and resolves deadlocks by rolling back the statement associated with the transaction that detects the deadlock.

Resolving Deadlock:

1) Ask the session getting deadlock error ORA-00060 to issue either **commit** or **rollback**.

2) Ask the waiting session to kill the SQL/transaction.

-> The following script can be used to identify deadlocks in the database.

-> The query depends upon objects that are created by the script

\$ORACLE_HOME/rdbms/admin/dbmslock.sql

Avoid deadlock:

->if you want to avoid the deadlock, try to use the clause skip locked.

(D) Attempt the following.

(05)

1) Write a note on View.

Ans. ->A view is a virtual table that stores only definition in the form of SQL statement.

-> It is used to hide information from other user.

Feature of view:

-> Creating view does not take any storage space in oracle.

-> The maximum number of columns that can be defined in a view is 1000.

-> When new records are inserted/updated/deleted in base table, the changes are automatically reflected in the view.

-> View once defined, can be used in Joins and Sub-query also.

Types of view:

1) Read only view

2) Updateable view

Syntax of view :-

```
CREATE [OR REPLACE] [FORCE] VIEW<View Name>
AS
SELECT<Fields> FROM <Table>WHERE <condition >
```

EXAMPLE:

1) Read only View:

Create or replace view vemp as

Select ename,sal from emp where sal>2000

With read only;

Insert into vemp values('SCOTT',3000);

2) Updatable View:

Create or replace view vemp as

Select ename,sal from emp where sal>1000;

Insert into vemp values('SCOTT',3000);

Q.7(a) write the following.

(1) Give full form of PL/SQL.

Ans. PL/SQL stands for Procedural Language/SQL

Or

Procedural Language extensions to SQL.

(2) Code blocks starts with a _____ section.

Ans. Declaration section

(3) A _____ object holds procedure, functions, variable, constants, cursors and exceptions.

Ans. Package

(4) An appropriate action is taken against the error condition known as _____

Ans. Exception block

(b) Answer in brief.

(1) What is %Rowtype?

Ans.

- Using %ROWTYPE attributes, a record variable can be declared implicitly based the structure of a table or query
- In implicit defined record, we don't have to describe each field in the record definition
- It mean oracle select data type automatically for declaring %ROWTYPE attribute
- Example :-
 - salary emp%rowtype

(c) Answer in detail.

(1) Explain a PL/SQL block structure.

Ans. PL/SQL block structure :

- Any language is useless if it has no programming power.
- By writing simple SQL statement, we cannot fulfill all our requirements.

- As result PL/SQL is required for the purpose of programming.
- PL/SQL stands for Procedural Language/SQL
- PL/SQL is bloc structure language. Each block divided into three sections
 - Declaration Section (Optional)
 - Execution Section
 - Exception Section (Optional)

Declaration Section :

- This is section is start with keyword DECLARE
- Variables and constants are declared, initialized within this section.
- You must have to declare variables and constants in declarative block before using them in procedural statement.
- Declare Variables and Assigning values: You can define variable name, data type of a variable and its size.
- Data type can be: CHAR, VARCHAR2, DATE, NUMBER, etc
- This section is optional if no variables or constraints to declare

Execution Section :

- This section is mandatory in PL/SQL
- This section begins with keyword BEGIN
- This section can have another PL/SQL block inside it
- Execution Section contains SQL – PL/SQL executable statements
- This section contains conditional statements (if...else), looping statements (for, while) and Branching Statements (goto) etc.

Exception Section:

- This is an Optional section
- This section start with keyword EXCEPTION
- PL/SQL easily detects user defined or predefined error condition.
- PL/SQL is famous for handling errors in smart way by giving suitable user friendly messages.
- Errors can be rise due to wrong syntax, bad logical or not passing a validation rules.

syntax :

DECLARE

<variable / constant declaration>

BEGIN

<statements>

```
EXCEPTION
    <statements>
END;
```

Example:

```
declare
    nam emp.ename%type;
begin
    nam:='&nam';
    select ename into nam from emp where ename=nam;
    dbms_output.put_line('employee found' || nam);
exception
    when no_data_found then dbms_output.put_line('emp not
found');
end;
```

(d) Answer the following

(1) write a note on Package.

Ans. Package :

- A package is an object, which holds other objects like procedure, functions, cursor, exception etc. within it
- Components of Package :-

Package specification

Package body

○ Package specification :-

- Contains Name of the package
- Contains declaration of Procedure, Function, Cursors, Exception etc.
- The objects declared in a package header are visible outside the package
- It does not contain any code for procedures / functions
- Object must be declared in this section before it is referenced

○ Syntax :

CREATE [OR REPLACE] PACKAGE <package name>

{AS / IS }

Public_Variable_declarations

Public_Exception_declarations

Public_Cursor_declarations

Function_specifications

Procedure_specifications

END [package name];

Package Body :

- Package body contains the definition of public objects that are declared in the specification
- Package body can also have other objects, which are private to the package
- If package header does not contain any procedure / functions then, package body is optional
- Private objects are defined in package body only and they are not visible / accessible outside the package.
- The variables declared in the package specification can be accessed by any procedure / function within the Package
- Package object referred as

EXECUTE <package Name>.<Object Name>

or

CALL <package Name>.<Object Name>

PACKAGE BODY SYNTAX :-

CREATE [OR REPLACE] PACKAGE BODY <package name>

{AS / IS }

Private_Variable_declarations

```
Private_Exception_declarations
Private_Cursor_declarations
Function_specifications
Procedure_specifications
END [package name];
```

Example:

-----package specification-----

create or replace package pack as

procedure pro1(n1 number,n2 number,n3 number);

procedure leap (y number);

end pack;

-----package body-----

create or replace package body pack as

procedure pro1(n1 number,n2 number,n3 number) as

vsum number(10);

begin

vsum:=n1+n2+n3;

dbms_output.put_line('sum: ' || vsum);

**dbms_output.put_line('greatest value:
' || greatest(n1,n2,n3));**

end pro1;

```
procedure leap(y number)as
begin
    if mod(y,4)=0 then
        dbms_output.put_line(y || ' is a leap year');
    else
        dbms_output.put_line(y || ' is not a leap year');
    end if;
end leap;
end pack;
```

```
-----execute package-----
execute pack.pro1(10,20,30);
execute pack.leap(2024);
```

Q.8 (a) Answer the following

(1) The _____ attribute to declare variables bases on definitions of columns in a table.

Ans. %type

(2) The _____ statement causes the loop to skip the remainder of its body and immediately retest its condition prior to reiterating.

Ans. Continue

(3) The oracle engine opened a cursor for its internal processing it is known as _____.

Ans. implicit cursor

(4) A table within a table is known as _____.

Ans. nested table

(b) Answer in brief.

(1) What is explicit cursor?

Ans. A cursor can also be opened for processing data through a PL/SQL block such a user-defined cursor known as explicit cursor

Syntax:

CURSOR cursor_name IS select_statement;

(c) Answer in detail.

(1) Explain Trigger.

Ans.

TRIGGER :

- Triggers are the programs that are executed automatically in response to a change in the database.
- Oracle allows special types of procedures that are automatically executed when events like INSERT/ UPDATE/ DELETE occurs. These event procedures called “DATABASE TRIGGERS”
- You may not use COMMIT, ROLLBACK and SAVEPOINT statement within trigger blocks.
- Types of triggers
 - Before update statement level
 - After insert row level
 - Before delete row level
- Before update statement level
 - In this type, trigger fires when any kind of updates occur
 - For ex. If you write query of update and this type of trigger fired on your table the changes occur before update any data
- After insert row level
 - In this type, trigger fires when row insert in database table
 - For ex. If you insert any data in your database table and trigger is fired in your table then after inserting row trigger run
- Before delete row level

- In this type, trigger fires before delete any record in table row
- For ex. If you are delete any records in your database table this trigger fires automatically and make changes as per trigger programmed

Syntax

```

CREATE [OR REPLACE] TRIGGER <name>
[BEFORE / AFTER / INSTEAD OF]
[DELETE / INSERT / UPDATE [OF <col1>,<col2>..]]
ON <table / view name>
FOR EACH ROW [WHEN <condition>]
DECLARE
    <VAR>
BEGIN
    PL/SQL STATEMENT
EXCEPTION
    EXCEPTION
END;
```

TRIGGER PREDICATES :

- When the trigger is created for more than one DML operation, one can use conditional predicates within the trigger body to

execute specific blocks of code, depending on the types of statement that fires the trigger.

- Inserting :
TRUE if the triggering statement is an INSERT
- Deleting:
TRUE if the triggering statement is an DELETE
- Updating:
TRUE if the triggering statement is an UPDATE

:OLD AND :NEW

- The :new refers to a record containing the new, post change data.
- The :old refers to a record containing the old, pre-change data.
- The trigger can have visibility to both old and new values.
- :NEW can be modified only in a before row-level trigger, and :OLD can never be modified but only read from

- Insert:
:old :-

Undefined all fields are null

:new :-

Values that will be inserted when the statement is complete

- Update:

:old :-

Original values for the row before update

:new :-

New values that will be updated when the statement is complete

- Delete:

:old :-

Original values before the row is deleted

:new :-

Undefined all fields are null

Example:

create or replace trigger msg

before insert or delete or update on stud

for each row

begin

if inserting then

dbms_output.put_line('Record inserted in stud');

elsif deleting then

dbms_output.put_line('Record deleting in stud');

elsif updating then

dbms_output.put_line('Record updating in stud');

end if;

end;

(d) Answer the following.

(1) Difference between Procedure V/S Function.

Function	Procedure
Functions always return a value after the execution of queries.	The procedure can return a value using “IN OUT” and “OUT” arguments.
In SQL, those functions having a DML statement can not be called from SQL statements. But autonomous transaction functions can be called from SQL queries.	A procedure can not be called using SQL queries.
Each and every time functions are compiled they provide output according to the given input.	Procedures are compiled only once but they can be called many times as needed without being compiled each time.
A Function can not return multiple result sets.	A procedure is able to return multiple result sets.
The function can be called using Stored Procedure.	While procedures cannot be called from function.
A function used only to read data.	A procedure can be used to read and modify data.
The return statement of a function returns the control and function’s result value to the calling program.	While the return statement of the procedure returns control to the calling program, it can not return the result value.

The function does not support try-catch blocks.	Procedure supports try-catch blocks for error handling.
A function can be operated in the SELECT statement.	While it can't be operated in the SELECT statement.
Functions do not permit transaction management.	It allows transaction management.
In functions, we can use only a table variable. Temporary tables can not be created in function.	In procedures, we can use temporary tables or table variables to store temporary data.

Q.9 (A) Answer The Following.

(1) Give Full Form of SGA.

=> System Global Area

(2) Give Full Form of SMON.

=> System MONiter

(3) Give Full Form of DBWn.

=> DataBase Writer Process

(4) Give Full Form of RBS.

=> Rollback Segment

(B) Answer in Brief.

(1) What is Control File?

=> The control file of database is binary file that contains a great deal

of database information The control file contains name, data about the database log file.

Oracle allows you to maintain duplicate copies with the help of control files.

Its good if more then one copies stored in different disks to protect your database information.

(C) Answer in Detail.

(1) Explain Oracle Block.

Ans. => PL/SQL block structure :-

- ❑ Any language is useless if it has no programming power.
- ❑ By writing simple SQL statement, we cannot fulfill all our requirements.
- ❑ As result PL/SQL is required for the purpose of programming.
- ❑ PL/SQL stands for Procedural Language/SQL.
- ❑ PL/SQL is bloc structure language. Each block divided into three sections.
- ❑ Declaration Section (Optional).
- ❑ Execution Section.
- ❑ Exception Section (Optional).

Declaration Section :-

- ❑ This section starts with the keyword DECLARE.
- ❑ Variables and constants are declared, initialized within this section.
- ❑ You must have to declare variables and constants in a declarative block before using them in a procedural statement.
- ❑ Declare Variables and Assigning values: You can define variable name, data type of a variable and its size.
- ❑ Data type can be: CHAR, VARCHAR2, DATE, NUMBER, etc.
- ❑ This section is optional if no variables or constraints to declare.

EXECUTABLE SECTION :-

- ❑ This section is mandatory in PL/SQL.
- ❑ This section begins with the keyword BEGIN.
- ❑ This section can have another PL/SQL block inside it.
- ❑ Execution Section contains SQL – PL/SQL executable statements.
- ❑ This section contains conditional statements (if...else), looping

statements (for, while) and Branching Statements (goto) etc.

EXCEPTION SECTION :-

- This is an Optional section.
- This section start with keyword EXCEPTION.
- PL/SQL easily detects user defined or predefined error condition.
- PL/SQL is famous for handling errors in smart way by giving suitable user friendly messages.
- Errors can be rise due to wrong syntax, bad logical or not passing a validation rules.

Plsql block structure syntax :-

Syntax

■ DECLARE

■ <variable / constant declaration>

■ BEGIN

■ <statements>

■[EXCEPTION

■<statements>]

■END;

(D) Answer the Following.

(1) Write a Note on Managing Automated Database Maintenance Tasks.

=> Managing Automated Database Maintenance Task in Oracle

- This is to refer the process of using Oracle Database's built-in automation features and tools to efficiently and proactively maintain the health, performance, and reliability of an Oracle database.
- Oracle provides a range of automated tasks and features to simplify routine maintenance activities that database administrators (DBAs) typically perform manually.
- **These tasks include:**

Backup and Recovery:

- Oracle offers tools like Recovery Manager (RMAN) to automate database backups, ensuring that data can be recovered in case of

a failure.

Patching and Upgrades:

- Oracle provides automated patching and upgrade solutions, such as Patch Set Updates (PSUs) and Critical Patch Updates (CPUs), to keep the database software up-to-date with the latest security and performance enhancements.

Statistics Gathering:

- Regularly collecting and refreshing database statistics is crucial for query optimization. Oracle can automatically gather statistics on database objects.

Performance Monitoring and Tuning:

- Oracle's Automated Database Diagnostic Monitor (ADDM) analyzes database performance and provides recommendations for improving it.

Space Management:

- Automated features help manage tablespace, control file, and redo log file space.

Task Scheduling:

- ❑ Oracle allows DBAs to schedule routine maintenance tasks, making it easier to execute them during non-peak hours.

Alerts and Notifications:

- ❑ Automated systems can generate alerts and notifications when predefined conditions or thresholds are met, allowing for proactive problem resolution.

Resource Management:

- ❑ Oracle Database can automatically allocate and manage system resources to optimize performance for various workloads.

Security Updates:

- ❑ Automatic security updates help protect the database against vulnerabilities.

Q.10 (A) Answer The Following.

(1) Give Full Form Of PGA.

=> **P**rogram **G**lobal **A**rea

(2) Give Full Form Of PMON.

=> **P**rocess **MON**iter

(3) Give Full Form Of CKPT.

=> **C**heck **P**oint **P**rocess

(4) Give Full Form Of LGWR.

=> **L**o**G** **W**Riter **P**rocess

(B) Answer in Brief.

(1) What is Rollback Segment.

=> In the ORACLE database, the rollback segment is a structure that stores previous data during transactions.

- To support the rollback facility we have a rollback segment in the ORACLE database.

(C) Answer in Detail.

(1) Explain Tablespace.

=> The Oracle tablespace and how Oracle uses tablespaces to logically store the data in the database.

What is an Oracle Tablespace?

- Oracle divides a database into one or more logical storage units called tablespaces.
- Each tablespace consists of one or more files called datafiles. A datafile physically stores the data objects of the database such as tables and indexes on disk.
- In other words, Oracle logically stores data in the tablespaces and physically stores data in datafiles associated with the corresponding tablespaces.

By using tablespaces, you can perform the following operations :-

- Control the storage size allocated for the database data.

- Grant specific space quotas to the database users.
- Control the availability of data by taking tablespaces online or offline (more on this later).
- Improve the performance of the database by allocating data storage across devices.
- Perform partial database backup or recovery.

Read-Only Tablespaces :-

- The read-only tablespaces allow Oracle to avoid performing backup and recovery of large, static parts of a database. Because Oracle doesn't update the files of a read-only tablespace, you can store the files on the read-only media.
- Oracle allows you to remove objects such as tables and indexes from a read-only tablespace. However, it does not allow you to create or alter objects in a read-only tablespace.

- When you create a new tablespace, it is in the read-write mode. To change a tablespace to read-only tablespace, you use the ALTER TABLESPACE command with the READ ONLY option.

More on Oracle Tablespaces :-

- **Create tablespace** – show you how to create a new tablespace in the database.
- **Drop tablespace** – describe the steps of removing a tablespace from the database.
- **Extend tablespace** – how to extend the size of a tablespace.
- **Temporary tablespace** – manipulate temporary tablespace effectively.
- **Tablespace group** – how to use the tablespace group more effectively to optimize internal Oracle operations.

(D) Answer the Following.

(1) Write A Note On Oracle Scheduler Concept.

=> Oracle Scheduler Concepts refer to the fundamental ideas and components that form the basis of Oracle Scheduler, a powerful job

scheduling and automation tool within the Oracle

Database. Oracle Scheduler is designed to help automate and manage a wide range of tasks, jobs, and processes within the database environment.

Job :-

- A job is a specific task or unit of work that you want to automate using Oracle Scheduler. Jobs can include activities like running SQL scripts, PL/SQL procedures, external scripts, or data transformations. Jobs are at the core of Oracle Scheduler.

Program :-

- A program defines what a job does. It specifies the executable and arguments required to perform a particular task. Programs can include stored procedures, executable files, or scripts. A job references a program to execute a specific task.

Schedule :-

- A schedule defines when and how often a job should run. It specifies the frequency, start time, end time, and recurrence pattern for executing a job. Schedules allow you to control the timing of job execution.

Job Class :-

- Job classes are used to categorize and manage jobs. You can assign jobs to specific job classes based on their characteristics or priorities. Job classes help in resource allocation and prioritization of job execution.

Window :-

- A window is a specific time interval during which jobs are allowed to run. Windows are often used to control when maintenance tasks or resource-intensive jobs can be executed, ensuring they don't interfere with critical workloads.

Dependency :-

- Dependency defines the relationship between jobs. You can specify that one job should only run after another job has completed successfully. Dependencies enable you to create job chains and manage complex workflows.

Scheduling Jobs with Oracle Scheduler

- "Scheduling Jobs with Oracle Scheduler" involves using Oracle Scheduler, a feature within the Oracle Database, to automate and manage the execution of various tasks or jobs at specified times

or under specific conditions.

Creating Jobs :-

- To schedule jobs with Oracle Scheduler, you first need to define the jobs you want to automate. Jobs can include tasks like running SQL scripts, PL/SQL procedures, or external scripts. These jobs are encapsulated as objects within the database.

Defining Programs :-

- For each job, you need to specify a program that defines what the job does. The program specifies the executable or procedure to run, as well as any required arguments or parameters. Programs can be created within the database or reference external executables.

Setting Schedules :-

- Once you have defined jobs and programs, you need to create schedules that determine when and how often these jobs should run. Schedules can be configured to execute jobs at specific times of the day, week, or month, and they can include options like specifying start and end times or recurring patterns.

Monitoring and Notifications :-

- Oracle Scheduler provides tools for monitoring the execution of scheduled jobs. You can view job status, logs, and performance metrics. Notifications can be configured to alert administrators when specific events or job outcomes occur.

Job History :-

- Oracle Scheduler maintains a history of job executions, allowing administrators to track the progress of jobs, view job status, and diagnose problems. This historical data can be valuable for auditing and troubleshooting purposes.

Security and Access Control :-

- Access control mechanisms ensure that only authorized users can create, modify, or execute scheduled jobs. This helps protect sensitive data and ensures that job scheduling is secure.

Job Management :-

- Administrators are responsible for creating, modifying, and deleting jobs. This includes defining the job type (e.g., SQL script, PL/SQL procedure), setting job attributes, specifying the program to execute, and assigning the job to schedules.

Programs and Executables :-

- Administrators manage programs and executables associated with jobs. This involves defining the programs, specifying executable files or PL/SQL procedures, and configuring program attributes.

Schedules :-

- Administering Oracle Scheduler involves the creation, modification, and deletion of schedules. Administrators determine when jobs should run by defining schedules, specifying start and end times, and configuring recurring patterns.

Dependency Management :-

- Administrators establish dependencies between jobs when needed. Dependencies define the order in which jobs are executed and ensure that specific conditions are met before a job starts.

Window Configuration :-

- Windows, which specify time intervals during which jobs are allowed to run, are set up and managed by administrators. Windows help control job execution during different time frames,

such as peak and off-peak hours.

Resource Allocation :-

- Resource plans and job classes are configured to manage resource allocation to jobs. Administrators ensure that critical jobs receive adequate system resources while preventing resource contention.

